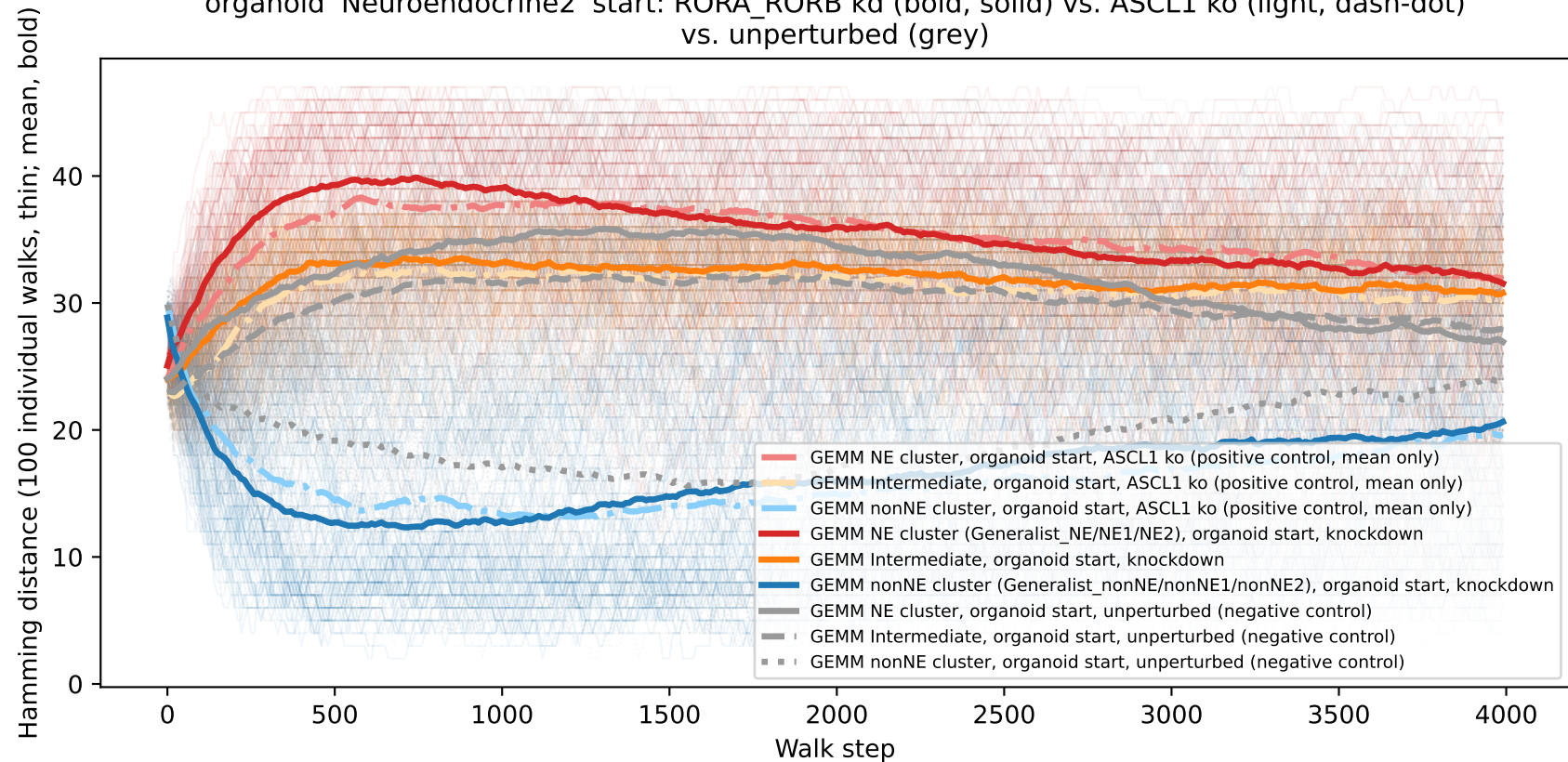


Distance to archetype clusters over time  
organoid 'Neuroendocrine2' start: RORA\_RORB kd (bold, solid) vs. ASCL1 ko (light, dash-dot)  
vs. unperturbed (grey)



Wilcoxon signed-rank, step of closest approach: Arc\_1 (Intermediate) vs. nonNE cluster (n=100 walks)  
RORA\_RORB kd: mean step 690 vs. 1330; Arc\_1 first in 81% of walks;  $p=5.6e-05$   
ASCL1 ko: mean step 774 vs. 1521; Arc\_1 first in 78% of walks;  $p=2.9e-05$   
Unperturbed: mean step 1125 vs. 1649; Arc\_1 first in 68% of walks;  $p=2.5e-02$